Microgravity Fire Safety Parameter Catalog

Project: Flame in Freefall - AI-Powered Fire Safety Insights from Microgravity Combustion Data

Purpose: Canonical parameter ontology, dataset discovery guide, and comparison framework

Version: 1.0

Last updated: 2026-10-01

Status: Editable research document - not a flight-safety standard

Executive summary

NASA microgravity combustion data is distributed across investigation pages, experimental tables, sensor logs, videos, images, reports, and publications. The same physical idea is often represented by different names, units, or file structures. This catalog provides one consistent vocabulary so that data can be found, normalized, compared, ranked, and explained without erasing scientifically important differences.

The catalog separates every record into three layers:

Layer	Meaning	Examples
Controlled inputs	Conditions selected or imposed during a test	gravity, pressure, oxygen, airflow, material, geometry, ignition
Measured outputs	What happened during or after combustion	ignition delay, flame spread, temperature, smoke, extinction
Evidence context	Why a result can or cannot be trusted and compared	instrument, calibration, source, uncertainty, processing, domain shift

The recommended storage hierarchy is:

```
Investigation
  -> Experiment or test
      -> Controlled inputs
      -> Measured outputs
      -> Evidence context
      -> Mission interpretation (derived, never treated as raw data)
```

The most important design rule is: **search by scientific concept, compare through canonical units, and retain the original NASA evidence.**

1. How to use this catalog

Use the catalog for four related tasks:

1. **Dataset discovery:** tag every investigation by the parameter groups it contains.
2. **Schema normalization:** map raw column names and metadata to canonical parameter names.
3. **Scientific comparison:** compare only records that pass explicit compatibility checks.
4. **Decision support:** produce mission-relevant evidence cards that distinguish known, inferred, and unknown information.

For each imported value, preserve both the original and normalized representation:

```
{
  "parameter": "flow_velocity",
  "value": 20.0,
  "unit": "cm/s",
  "original_name": "Air Speed",
```

```
  "original_value": 0.2,
  "original_unit": "m/s",
  "measurement_role": "controlled_input",
  "source_test_id": "SAFFIRE-I-S1",
  "source_file": "telemetry.csv",
  "uncertainty": 0.8,
  "uncertainty_unit": "cm/s",
  "processing_method": "unit_conversion",
  "confidence": 0.96
}
```

Never overwrite the original NASA field or value during normalization.

2. Sorting system

Each parameter receives four independent labels. This avoids forcing one parameter into a single rigid folder.

Sorting axis	Allowed values	Purpose
Evidence role	controlled_input, measured_output, derived_output, context_metadata	Separates causes, observations, calculations, and metadata
Physical domain	gravity, atmosphere, flow, material, ignition, combustion, thermal, smoke, detection, suppression	Supports scientific filtering
Data shape	scalar, category, vector, time_series, image, video, spectrum, document	Selects processing and visualization methods
Evidence level	raw, calibrated, processed, reported, modeled, AI_extracted	Prevents derived values from being mistaken for measurements

2.1 Parameter naming convention

- Use lowercase `snake_case`.
- Use one physical meaning per canonical name.
- Add a suffix only when it prevents ambiguity: `_rate`, `_fraction`, `_time`, `_temperature`, `_velocity`.
- Do not embed units in the parameter name.
- Store units in a separate field.
- Use controlled vocabularies for categories.
- Use identifiers rather than filenames as primary keys.

2.2 Required record identifiers

Every record should be traceable through this key chain:

```
investigation_id -> experiment_id -> test_id -> observation_id
```

3. Group index

Code	Group	Role	Primary search question
G01	Provenance and identity	Context	Where did this evidence come from?
G02	Mission and operational scenario	Context/derived	Where might the evidence be applied?
G03	Gravity and platform dynamics	Input/context	What gravity environment existed?
G04	Atmosphere and ambient conditions	Input	What gas environment surrounded the fire?
G05	Flow, ventilation, and confinement	Input	How were oxidizer and products transported?
G06	Material and fuel properties	Input	What was burning?
G07	Specimen geometry and configuration	Input	In what form and arrangement did it burn?
G08	Ignition and external exposure	Input/output	How was combustion initiated?
G09	Flame propagation and burning behavior	Output	How did the fire develop?
G10	Thermal and energy behavior	Output	What heat and temperature were produced?
G11	Smoke, aerosol, and gas products	Output	What products and crew hazards were generated?
G12	Detection and alarm performance	Output	How quickly and reliably was the fire detected?
G13	Suppression, extinction, and recovery	Input/output	How was the fire controlled and what remained?
G14	Instrumentation and synchronization	Context	How were observations collected?
G15	Data quality, uncertainty, and comparability	Context/derived	Can this evidence be trusted and compared?
G16	Derived safety intelligence	Derived	What decision-relevant conclusion is supported?

4. G01 - Provenance and identity

This group makes every observation findable, citable, and reproducible.

Canonical parameter	Type	Common aliases	Description
investigation_id	string	PSI ID, study ID, project ID	Stable repository-level identifier
investigation_name	string	project, campaign, investigation	Full investigation title
investigation_acronym	category	acronym, short name	BASS, FLEX, SAFFIRE, SAME, ACME, and similar
experiment_id	string	experiment, configuration ID	Identifier below investigation level
test_id	string	run ID, burn number, trial, sample test	Smallest repeatable experimental event
observation_id	string	measurement ID, record ID	Unique observation identifier
platform	category	flight platform, facility	ISS, Cygnus, drop tower, aircraft, ground chamber
facility	string	apparatus, rack, chamber	CIR, MSG, MDCA, wind tunnel, test cell
test_date	date/time	run date, GMT, timestamp	When the test occurred
principal_investigator	string	PI, research lead	Responsible investigator

Canonical parameter	Type	Common aliases	Description
sponsoring_agency	string	agency, sponsor	NASA or partner organization
nasa_center	string	center, institution	Responsible NASA center
dataset_version	string	revision, release	Version of the source dataset
source_doi	string	DOI, persistent ID	Stable citation identifier
source_url	URL	landing page, download URL	Source location
source_file	string	filename, asset	File containing the observation
source_file_version	string	file revision	Version of the individual file
data_level	category	raw, processed, analyzed	Maturity of the data
license	category	rights, usage terms	Reuse license
related_investigation	list	related PSI, predecessor	Linked investigations

Recommended data levels: `raw`, `calibrated`, `processed`, `derived`, `modeled`, `reported_summary`.

5. G02 - Mission and operational scenario

Mission fields describe a target use case. They must not be presented as if they were measured during a laboratory test.

Canonical parameter	Type	Example values	Notes
destination	category	ISS, Gateway, Moon, Mars, transit	Mission destination or environment
mission_phase	category	launch, cruise, docking, surface, EVA prep	Operational phase
habitat_type	category	spacecraft, module, lander, laboratory	Enclosure type
operational_state	category	crewed, uncrewed, dormant, emergency	System state
mission_duration	duration	days, months, years	Exposure and logistics horizon
crew_count	integer	0 to mission maximum	Number of exposed crew
abort_capability	category	immediate, delayed, unavailable	Consequence modifier
response_delay	duration	seconds or minutes	Expected response time
fire_location	category	cabin, rack, avionics, storage, duct	Operational location
system_criticality	ordinal	low, medium, high, mission critical	Consequence classification
available_suppressant	list	CO2, water mist, inert gas	Available response resources
post_fire_filtration	category	none, limited, full	Cleanup capability
repairability	category	replaceable, repairable, nonrepairable	Recovery constraint

Mission scenario records should be linked to experimental evidence through a separate `scenario_evidence_link` table containing similarity and domain-shift values.

6. G03 - Gravity and platform dynamics

Canonical parameter	Role	Canonical unit/value	Common aliases
gravity_regime	input/context	microgravity, lunar, Martian, Earth, variable	micro-g, zero-g, low-g, partial-g
gravity_level	input	fraction of Earth g	g-level, acceleration level
gravitational_acceleration	input	m/s^2	acceleration, effective gravity
gravity_vector	input	3-element vector	acceleration vector
gravity_direction_relative_to_sample	input	category/vector	sample-gravity orientation
reduced_gravity_duration	context	s	low-g time, micro-g duration
residual_acceleration	context	fraction of Earth g	residual-g, g-jitter
vibration_level	context	acceleration spectrum	platform vibration
rotation_rate	context	rad/s	centrifuge speed, angular velocity
gravity_gradient	context	1/s^2 or descriptive	centrifuge gradient

Do not collapse microgravity, lunar gravity, Martian gravity, and reduced-pressure Earth tests into one `space` category.

7. G04 - Atmosphere and ambient conditions

Canonical parameter	Role	Preferred unit	Common aliases
total_pressure	input	kPa	ambient pressure, cabin pressure, chamber pressure
oxygen_mole_fraction	input	fraction	O2%, oxygen level, XO2
oxygen_partial_pressure	derived/input	kPa	pO2, oxygen pressure
nitrogen_mole_fraction	input	fraction	N2 fraction
carbon_dioxide_mole_fraction	input	fraction	initial CO2, background CO2
suppressant_mole_fraction	input	fraction	diluent fraction, agent percentage
suppressant_species	input	category	CO2, He, N2, SF6, other
gas_composition	input	structured mixture	atmosphere, oxidizer composition
ambient_temperature	input	K	cabin temperature, initial gas temperature
relative_humidity	input	%	RH, humidity
gas_density	input/derived	kg/m^3	ambient density
initial_oxygen_concentration	input	mol/m^3	initial O2 concentration
initial_contaminant_level	input	ppm or mg/m^3	background gas level

Important distinction: oxygen mole fraction and oxygen partial pressure are related but not interchangeable. Retain both when pressure varies.

8. G05 - Flow, ventilation, and confinement

Canonical parameter	Role	Preferred unit/value	Common aliases
flow_condition	input	quiescent, forced, mixed	still air, ventilation state
flow_velocity	input	cm/s	airspeed, forced flow, ventilation velocity
flow_direction	input	vector/category	airflow direction
flow_configuration	input	concurrent, opposed, cross-flow	downstream/upstream configuration
volumetric_flow_rate	input	L/min	airflow rate, fan flow
mass_flow_rate	input	g/s	oxidizer flow, gas feed rate
turbulence_intensity	input/context	%	turbulence level
reynolds_number	derived	dimensionless	Re
residence_time	derived	s	reactant residence time
damkohler_number	derived	dimensionless	Da
fan_state	input	on, off, speed level	blower state
duct_cross_section	context	m^2	tunnel area, flow area
chamber_volume	context	m^3	enclosure volume
hydraulic_diameter	derived/context	m	duct characteristic length
wall_distance	context	mm	distance to boundary
confinement_ratio	derived	dimensionless	blockage or confinement ratio
inlet_location	context	coordinates	vent inlet position
outlet_location	context	coordinates	vent outlet position

Flow velocity, volumetric flow, and mass flow are different measurements. Convert only when geometry and thermodynamic conditions are available.

9. G06 - Material and fuel properties

Canonical parameter	Role	Preferred value/unit	Common aliases
material_name	input	controlled name	sample, fuel, specimen material
material_trade_name	input	text	commercial product name
material_class	input	polymer, fabric, foam, wire, liquid, gas	fuel class
chemical_family	input	PMMA, cellulose, hydrocarbon, alcohol	chemistry group
material_composition	input	structured mixture	constituents, blend
substrate_material	input	controlled name	backing, support
coating_material	input	controlled name	surface treatment
material_density	input	kg/m^3	bulk density
porosity	input	fraction	void fraction
moisture_content	input	mass fraction	water content
thermal_conductivity	input	W/(m K)	conductivity
specific_heat_capacity	input	J/(kg K)	heat capacity
emissivity	input	fraction	surface emissivity
heat_of_combustion	input	MJ/kg	effective heat of combustion
pyrolysis_temperature	input/model	K	decomposition temperature
ignition_temperature	input/model	K	critical surface temperature
material_condition	input	new, aged, contaminated, damaged	specimen condition
melting_behavior	output/category	none, softening, melting, dripping	melt response
char_behavior	output/category	none, thin, thick, expanding	charring/intumescence
spacecraft_relevance	context	flight material, analog, idealized	application realism

`material_name`, `material_class`, and `chemical_family` must remain separate so users can search at different levels of specificity.

10. G07 - Specimen geometry and configuration

Canonical parameter	Role	Preferred unit/value	Common aliases
sample_shape	input	sheet, slab, rod, wire, droplet, sphere	geometry, specimen form
sample_length	input	mm	length
sample_width	input	mm	width
sample_thickness	input	mm	depth, gauge, film thickness
sample_diameter	input	mm	rod/wire/droplet diameter
initial_droplet_diameter	input	mm	droplet size, D0
sample_mass	input	g	initial mass
exposed_area	input/derived	cm^2	burnable area
sample_orientation	input	vector/category	orientation
orientation_to_flow	input	degrees/category	alignment with airflow
orientation_to_gravity	input	degrees/category	upward/downward geometry
sample_arrangement	input	continuous, discrete, layered, bundled	fuel layout
sample_spacing	input	mm	gap, separation
sample_count	input	integer	number of droplets/specimens
mounting_method	context	clamped, frame, free, supported	holder configuration
edge_condition	context	exposed, framed, sealed	boundary condition
adjacent_material	context	controlled name	propagation neighbor
distance_to_wall	context	mm	enclosure clearance

Terms such as `upward` and `downward` are defined relative to gravity. Terms such as `concurrent` and `opposed` are defined relative to forced flow. Do not use them as synonyms.

11. G08 - Ignition and external exposure

Canonical parameter	Role	Preferred unit/value	Common aliases
igniter_type	input	hot wire, spark, radiant, hot surface	ignition source
ignition_mode	input	piloted, spontaneous, electrical	ignition method
ignition_location	input	edge, center, upstream, downstream	igniter position
ignition_energy	input	J	spark/ignition energy
igniter_power	input	W	heater power
ignition_duration	input	s	heating time, exposure time
igniter_temperature	input	K	heater temperature
external_heat_flux	input	kW/m^2	radiant flux, imposed flux
ignition_area	input	mm^2	heated/contact area
ignition_success	output	true/false/partial	ignition outcome
ignition_delay	output	s	time to ignition
smolder_to_flame_transition	output	true/false/time	transition event
reignition_event	output	true/false	recurrence after extinction
reignition_delay	output	s	time to recurrence

Do not mix controlled exposure time with measured ignition delay.

12. G09 - Flame propagation and burning behavior

Canonical parameter	Role	Preferred unit/value	Common aliases
flame_presence	output	boolean/time series	flame detected
flame_front_position	output	mm or coordinates	leading edge position
flame_trailing_edge_position	output	mm or coordinates	trailing edge position
flame_centroid	output	coordinates	flame center
flame_length	output	mm	luminous length
flame_width	output	mm	luminous width
projected_flame_area	output	mm^2	image flame area, luminous area
estimated_flame_volume	derived	mm^3	reconstructed volume
flame_standoff_distance	output	mm	distance from fuel surface
flame_spread_direction	output	vector/category	propagation direction
flame_spread_rate	output	mm/s	spread velocity, propagation speed
flame_growth_rate	output	area/time or length/time	growth rate
burning_duration	output	s	flame lifetime
burning_rate	ambiguous	must be specialized	do not use without definition
mass_loss_rate	output	g/s	fuel consumption rate
regression_rate	output	mm/s	surface/droplet regression
droplet_d_squared_slope	output	mm^2/s	droplet burning constant
flame_luminosity	output	calibrated intensity	brightness
flame_color_metric	output	calibrated color index	color ratio
flame_stability	output	steady, oscillating, intermittent	stability class
oscillation_frequency	output	Hz	pulsation frequency
local_extinction_event	output	time/location	partial flame loss
extinction_event	output	true/false/time	flameout, quenching
extinction_time	output	s	time to extinction
extinction_mechanism	derived/reported	quenching, blowoff, burnout, suppression	cause
limiting_oxygen_condition	output	fraction and pressure	LOI, MOC, extinction oxygen
low_velocity_quenching_limit	output	cm/s	radiative/quenching boundary
high_velocity_blowoff_limit	output	cm/s	blowoff boundary
residual_glowing	output	boolean/duration	glowing after flame
cool_flame_event	output	boolean/time range	low-temperature combustion

Never store an unspecialized `burn_rate`. Determine whether the source means flame spread, mass loss, surface regression, or droplet diameter change.

13. G10 - Thermal and energy behavior

Canonical parameter	Role	Preferred unit	Notes
flame_temperature	output	K	Include location and sensor
gas_temperature	output	K	Include coordinates
surface_temperature	output	K	Fuel surface temperature
plume_temperature	output	K	Downstream combustion products
wall_temperature	output	K	Enclosure response
heat_release_rate	output/derived	kW	HRR; state calculation method
peak_heat_release_rate	derived	kW	Maximum HRR
total_heat_release	derived	kJ	Time-integrated HRR
radiative_heat_flux	output	kW/m^2	Radiometer measurement
convective_heat_flux	derived/output	kW/m^2	Convective transfer
total_heat_flux	output	kW/m^2	Combined heat flux
radiative_fraction	derived	fraction	Radiation/total energy
pyrolysis_rate	output/model	g/s	Fuel-generation rate
thermal_damage_area	output	cm^2	Heat-affected region
burnout_time	output	s	Fuel consumption endpoint

Every temperature and heat-flux value should link to sensor position, calibration, and timestamp.

14. G11 - Smoke, aerosol, and gas products

Canonical parameter	Role	Preferred unit/value	Common aliases
smoke_generation_rate	output	g/s or calibrated proxy	smoke production
soot_yield	output	g soot/g fuel	soot production factor
soot_volume_fraction	output	fraction	soot concentration
aerosol_number_concentration	output	particles/cm^3	particle count
aerosol_mass_concentration	output	mg/m^3	particle mass
particle_size_distribution	output	distribution	size spectrum
particle_mean_diameter	derived	nm or um	state averaging method
particle_median_diameter	derived	nm or um	state number/mass basis
particle_morphology	output/category	sphere, aggregate, chain, droplet	particle form
optical_density	output	1/m or dimensionless	smoke opacity
light_obscuration	output	%/m	visibility impact
visibility_distance	derived	m	visibility estimate
smoke_transport_velocity	output	cm/s	aerosol motion
smoke_arrival_time	output	s	time at detector/location
smoke_deposition	output	mg/m^2	surface/filter deposition
oxygen_consumption	output	mol or rate	decrease in O2
carbon_dioxide_concentration	output	ppm/fraction	CO2 product
carbon_monoxide_concentration	output	ppm	CO hazard
hydrogen_cyanide_concentration	output	ppm	HCN hazard
hydrogen_chloride_concentration	output	ppm	HCl hazard
hydrogen_fluoride_concentration	output	ppm	HF hazard
formaldehyde_concentration	output	ppm	cool-flame/toxicity product
post_fire_persistence_time	derived	s/min	cleanup burden

Always preserve concentration basis: mole fraction, mass concentration, or instrument signal.

15. G12 - Detection and alarm performance

Canonical parameter	Role	Preferred unit/value	Notes
detector_id	context	identifier	Links observations to device
detector_type	input/context	photoelectric, ionization, gas, thermal	Detection principle
detector_model	context	text	Manufacturer/model when public
detector_location	context	coordinates/category	Duct, chamber, wall
detector_orientation	context	vector/category	Relative to flow
detector_threshold	input	device-specific/calibrated	Alarm setting
detector_response_signal	output	time series	Raw detector output
detection_time	output	s	Time from ignition to alarm
detector_particle_range	context	nm or um	Sensitive size range
detector_gas_sensitivity	context	species/range	Gas detection coverage
false_positive_rate	output/validation	fraction	Reliability metric
false_negative_rate	output/validation	fraction	Missed-detection metric
monitored_volume	derived/context	m^3	Effective coverage
ventilation_dependency	derived/reported	low, medium, high	Transport dependency

Keep smoke production and detector performance separate. A smoke-rich fire may still evade a detector if transport or particle response is unfavorable.

16. G13 - Suppression, extinction, and recovery

Canonical parameter	Role	Preferred unit/value	Common aliases
suppressant_species	input	CO2, N2, He, SF6, water mist	suppression agent
suppressant_concentration	input	fraction	agent percentage
suppressant_mass	input	g	delivered mass
suppressant_volume	input	L	delivered volume
suppressant_flow_rate	input	g/s or L/min	discharge rate
discharge_duration	input	s	agent delivery time
discharge_direction	input	vector/category	nozzle direction
nozzle_distance	input	mm	distance to fire
mist_droplet_size	input	um	water-mist size
activation_mode	input	manual, automatic	trigger mode
activation_delay	input/output	s	ignition-to-discharge delay
flame_reduction_fraction	output	fraction	suppression effectiveness
spread_arrest_time	output	s	time to stop propagation
extinguishment_time	output	s	agent activation to extinction
extinguishment_success	output	boolean/partial	outcome
reignition_after_suppression	output	boolean/time	recurrence
residual_hotspot_temperature	output	K	hidden re-ignition risk
agent_toxicity_constraint	context	category	crew compatibility
equipment_compatibility	context	category	corrosion/electronics impact
filtration_load	derived	mass/time	cleanup burden
habitat_recovery_time	derived	duration	return-to-use estimate

17. G14 - Instrumentation and synchronization

Canonical parameter	Type	Examples
instrument_id	identifier	camera, thermocouple, radiometer, gas sensor
instrument_type	category	visible camera, IR camera, anemometer
instrument_location	coordinates	x/y/z or named position
sampling_frequency	Hz	sensor sampling rate
timestamp_reference	category	GMT, mission elapsed time, local clock
synchronization_offset	s	offset from experiment master clock
calibration_date	date	latest calibration
calibration_method	text/category	calibration procedure
measurement_range	structured	minimum and maximum
instrument_resolution	unit-specific	smallest resolvable value
sensor_response_time	s	dynamic response
camera_resolution	pixels	width x height
camera_frame_rate	frames/s	FPS
camera_exposure	s	shutter/exposure time
camera_gain	device-specific	sensor gain
camera_viewpoint	category/pose	side, front, oblique
camera_spectral_band	nm range	visible/IR/filter band
pixel_scale	mm/pixel	spatial calibration
lens_distortion_model	structured	calibration coefficients
region_of_interest	pixels	ROI bounds
compression_method	category	codec or image format
missing_frame_count	integer	video completeness

Image-derived measurements are comparable only after checking viewpoint, calibration, spectral response, exposure, and compression.

18. G15 - Data quality, uncertainty, and comparability

Canonical parameter	Type	Meaning
measurement_uncertainty	numeric	Instrument and measurement uncertainty
uncertainty_type	category	standard, expanded, estimated, unknown
calibration_status	category	valid, expired, unknown, not applicable
missing_data_fraction	fraction	Proportion of expected data missing
data_gap_intervals	list	Missing time intervals
replicate_count	integer	Repeated tests under equivalent conditions
extraction_method	category	reported, manual, algorithmic, AI-assisted
extraction_confidence	fraction	Confidence in extracted value
source_quality	ordinal	raw, calibrated, peer-reviewed, summary
processing_history	list	Ordered transformations
comparison_normalization	list	Unit/image/time normalization applied
comparability_level	category	direct, normalized, partial, contextual, unsuitable
condition_match_score	fraction	Similarity of experimental conditions
domain_shift_score	fraction	Difference between evidence and target scenario
extrapolation_distance	numeric/vector	Distance outside observed parameter space
out_of_distribution_flag	boolean	Target outside supported domain
conflicting_evidence_flag	boolean	Material disagreement among sources
evidence_count	integer	Supporting tests/records
evidence_independence	ordinal	Repeated files versus independent experiments
model_validation_status	category	unvalidated, partial, validated in domain

18.1 Comparability labels

Label	Meaning
Directly comparable	Same physical definition, compatible apparatus, units, and measurement method
Comparable after normalization	Differences can be corrected through documented calibration or conversion
Partially comparable	Some important conditions differ; comparison is qualitative or conditional
Contextual evidence only	Relevant background, but not suitable for numerical comparison
Not comparable	Definitions, instruments, or physical regimes are incompatible

18.2 Required comparison checks

Before ranking two tests as similar, check:

- Material identity and fuel form
- Sample geometry and scale
- Gravity regime and duration
- Total pressure and oxygen partial pressure
- Flow velocity and direction
- Ignition mode and external heating
- Confinement and wall effects
- Instrument definition and calibration
- Outcome definition

- Time reference and processing history

19. G16 - Derived safety intelligence

These parameters are produced by the application. They must never be labeled as direct NASA measurements.

Derived dimension	Primary supporting groups	Recommended display
ignition_susceptibility	G04, G06, G07, G08	evidence range plus confidence
propagation_hazard	G05, G06, G07, G09	separate spread and growth indicators
thermal_hazard	G09, G10	heat/temperature evidence
smoke_hazard	G05, G11	production plus transport
toxicity_hazard	G11	species-specific concentrations
detectability	G05, G11, G12	detection time and coverage
suppressibility	G04, G09, G13	extinction and agent evidence
reignition_hazard	G08, G09, G13	residual combustion evidence
mission_consequence	G02	consequence and response constraints
evidence_confidence	G01, G14, G15	transparent quality score
research_gap_severity	all groups	missing parameter combinations

Use a multidimensional safety profile rather than a single opaque risk score.

19.1 Known, inferred, unknown classification

Class	Definition
Known	Directly measured or explicitly reported for matching conditions
Inferred	Estimated from related evidence with similarity and uncertainty shown
Unknown	Insufficient or unsuitable evidence

20. Alias and synonym map

Raw or historical terms	Canonical parameter	Normalization warning
O2%, oxygen level, oxygen concentration, XO2	oxygen_mole_fraction	Confirm percent versus fraction
pO2, oxygen pressure	oxygen_partial_pressure	Do not substitute mole fraction alone
cabin pressure, ambient pressure, chamber pressure	total_pressure	Confirm absolute versus gauge pressure
airspeed, forced flow, ventilation velocity	flow_velocity	Confirm local versus nominal flow
airflow rate, fan flow	volumetric_flow_rate	Requires duct area to derive velocity
flame velocity, propagation speed, spread velocity	flame_spread_rate	Confirm direction and definition
burn rate	ambiguous	Split into spread, regression, mass-loss, or d-squared rate
sample depth, gauge, film depth	sample_thickness	Confirm geometry
micro-g, zero-g, low-g	gravity_regime	Also store numerical gravity level
flameout, quenched, extinguished	extinction_event	Preserve mechanism separately
LOI, MOC, extinction oxygen, O2 limit	limiting_oxygen_condition	Preserve test method and pressure
luminous area, segmented flame, flame image area	projected_flame_area	Preserve threshold and camera settings
smoke density, opacity	optical_density	Confirm measurement basis
detector alarm delay, response time	detection_time	Define reference event
suppressant percentage, diluent level	suppressant_concentration	Confirm mole, mass, or volume basis

21. Canonical unit policy

Quantity	Preferred unit	Preserve original?
Length	mm	Yes
Area	mm^2 or cm^2	Yes
Volume	cm^3 or m^3 depending scale	Yes
Time	s	Yes
Velocity	cm/s for low-speed flame tests	Yes
Mass	g	Yes
Mass rate	g/s	Yes
Temperature	K	Yes
Pressure	kPa absolute	Yes
Gas fraction	mole fraction from 0 to 1	Yes
Heat flux	kW/m^2	Yes
Heat release rate	kW	Yes
Particle diameter	nm or um	Yes
Gas concentration	ppm or mole fraction	Yes
Acceleration	fraction of Earth g and m/s^2	Yes

Conversions must store the conversion method and never discard the source representation.

22. Dataset-family coverage map

The current local catalog contains 24 PSI investigations, including 19 flight investigations and five related ground/modeling studies. It lists 22 experimental test tables and roughly 3,000 test rows. The following map identifies the strongest dataset family for each question.

Dataset family	PSI IDs in local catalog	Strong parameter groups	Best discovery question
FLEX / FLEX-2	PSI-69, PSI-68	G03-G05, G06-G08, G09-G11, G13-G15	How do liquid droplets burn, extinguish, and respond to suppressants?
Cool Flame Investigation	PSI-39 plus PSI-117/142 studies	G04-G05, G06, G09-G11, G15	When does visible extinction transition to cool-flame chemistry?
BASS / BASS-II	PSI-26, PSI-25 plus PSI-62	G03-G10, G13-G15	How do solid materials ignite, spread, and extinguish in low-speed flow?
SAFFIRE I-III	PSI-98, PSI-99, PSI-100	G03-G05, G06-G11, G14-G15	How do larger spacecraft-scale fires grow and affect the atmosphere?
SAME / SAME-R	PSI-102, PSI-101 plus PSI-115	G05, G06, G11-G12, G14-G15	Which smoke properties affect spacecraft detection?
DAFT / DAFT-2	PSI-47	G11-G12, G14-G15	Can aerosol measurement hardware operate reliably in microgravity?
SPICE	PSI-107 plus PSI-60	G04-G06, G09-G11, G14-G15	When do gaseous flames begin producing soot?
SLICE	PSI-106	G04-G05, G09-G10, G14	How do flame structure and liftoff depend on flow?
ACME BRE	PSI-20	G04-G10, G14-G15	Can gaseous burning emulate material burning-rate and extinction behavior?
ACME CLD	PSI-21	G04-G05, G09-G11, G14	How do coflow diffusion flames form soot and radiation?
ACME E-FIELD	PSI-22	G04-G05, G09-G10, G14	How do electric fields change laminar diffusion flames?
ACME s-Flame	PSI-23	G04-G05, G09-G10, G14	How do spherical diffusion flames respond and extinguish?
ACME Flame Design	PSI-10	G04-G06, G09-G11, G14	How do flame configuration and fuel/oxidizer streams change soot?
ACME CFI-G	PSI-159	G04-G06, G09-G11, G14	What governs gaseous cool-flame behavior?

22.1 Dataset selection by task

User task	Start with	Then add
Solid-material flame spread	BASS-II, BASS	SAFFIRE, SoFIE publications, ground comparison
Large-fire growth	SAFFIRE I-III	BASS-II, modeling reports
Liquid-fuel extinction	FLEX, FLEX-2	CFI and ground studies
Suppression effectiveness	FLEX	BASS suppression tests and reports
Smoke detection	SAME, SAME-R	DAFT, SAFFIRE gas/smoke measurements
Soot formation	SPICE, ACME CLD	Flame Design, SLICE
Mission scenario comparison	BASS-II plus SAFFIRE	NASA-STD-6001 context and published models
ML/video analysis	BASS-II and SAFFIRE video	FLEX image sequences after calibration review

23. Search and filtering architecture

The dashboard should guide users from mission question to evidence rather than asking them to know NASA investigation names.

```
Mission or application
  -> fire-safety question
      -> controlled conditions
          -> observed outcomes
              -> evidence quality
                  -> investigation and files
```

23.1 Recommended filters

1. Mission/destination
2. Question: ignition, spread, smoke, detection, suppression, extinction
3. Fuel state: solid, liquid, gas, aerosol/smoldering
4. Material class and specific material
5. Gravity regime
6. Pressure and oxygen partial pressure
7. Flow condition and velocity
8. Geometry and configuration
9. Measurement type
10. Data level
11. Comparability and confidence
12. Investigation and publication source

23.2 Example discovery query

```
Question: solid-material flame spread
Gravity: microgravity
Flow: forced, 10-20 cm/s
Material class: fabric
Outputs: spread rate, flame area, oxygen change
Evidence: raw or calibrated, video available
```

The query engine should return BASS/BASS-II and relevant SAFFIRE tests first, but explain geometry and scale differences.

24. Minimum viable prototype schema

Start with the fields below before expanding to the full catalog.

Section	Required fields
Identity	investigation_id, test_id, investigation_acronym, platform, source_doi, source_file, data_level
Gravity	gravity_regime, gravity_level, reduced_gravity_duration
Atmosphere	total_pressure, oxygen_mole_fraction, oxygen_partial_pressure, ambient_temperature
Flow	flow_condition, flow_velocity, flow_configuration, chamber_volume
Material	material_name, material_class, chemical_family, spacecraft_relevance
Geometry	sample_shape, sample_length, sample_width, sample_thickness or diameter, sample_orientation
Ignition	igniter_type, ignition_duration, ignition_location, ignition_success, ignition_delay
Flame	flame_spread_rate, projected_flame_area, flame_length, burning_duration, extinction_time
Smoke/gas	smoke indicator, oxygen change, carbon_dioxide concentration, carbon_monoxide when available
Suppression	suppressant_species, suppressant_concentration, extinguishment_success, reignition
Evidence	instrument_id, extraction_method, uncertainty, comparability_level, domain_shift_score

24.1 Prototype priority tiers

Tier	Include	Reason
Tier 1	identity, material, pressure/O2, airflow, geometry, spread/extinction, source	Necessary for credible comparison
Tier 2	synchronized video, temperature, gas measurements, suppression	Creates useful multimodal evidence
Tier 3	toxicity, detailed material properties, mission consequence, modeling	Valuable but dependent on sparse metadata

25. Recommended database structure

Use related tables rather than one extremely wide spreadsheet.

Table	Primary key	Contents
investigations	investigation_id	PSI-level metadata and provenance
experiments	experiment_id	Apparatus/configuration-level metadata
tests	test_id	Controlled inputs and specimen configuration
observations	observation_id	Long-format measured values
time_series	series_id	Sensor arrays and synchronized timestamps
media_assets	asset_id	Video/image metadata and calibration
instruments	instrument_id	Sensor definitions and locations
publications	publication_id	NTRS/DOI references
mission_scenarios	scenario_id	User-defined target conditions
evidence_links	link_id	Test-to-scenario similarity and domain shift
parameter_dictionary	parameter_id	Canonical names, aliases, types, units

25.1 Long-format observation model

Prefer:

```
test_id | time | parameter | value | unit | instrument_id | uncertainty
```

over a different wide-table schema for every investigation. Wide source tables may be retained in a raw staging layer.

26. Ranking and comparison logic

26.1 Evidence relevance

Rank evidence using visible components rather than an opaque AI score:

Component	Example comparison
Material similarity	same material > same chemical family > same broad class
Geometry similarity	same shape/thickness > scaled analog > different form
Gravity similarity	same gravity regime > validated correlation > extrapolation
Atmosphere similarity	distance in pressure and oxygen partial pressure
Flow similarity	velocity and concurrent/opposed configuration
Ignition similarity	same source, location, duration, and heat flux
Scale similarity	specimen and enclosure dimensions
Measurement quality	calibrated direct measurement > derived estimate > narrative report

26.2 Evidence confidence

Confidence should combine, but not hide:

- Source quality
- Calibration status
- Replicate count
- Missingness
- Measurement uncertainty
- Agreement among independent experiments
- Distance from the target scenario
- Validation status of any model used

26.3 Do not produce unsupported predictions

If a scenario lies outside the observed range of gravity, oxygen, pressure, material, flow, or geometry, mark it as `unknown` or `out_of_distribution`. A research-gap result is more valuable than a falsely precise prediction.

27. Data ingestion workflow

```
1. Register investigation and source files
2. Preserve raw tables and media unchanged
3. Profile raw column names, units, missingness, and types
4. Map raw names to canonical parameters
5. Validate unit conversions
6. Separate controlled inputs from measured outputs
7. Attach instruments, calibration, timestamps, and uncertainty
8. Compute only documented derived fields
9. Assign comparability metadata
10. Publish searchable evidence records with source links
```

27.1 Mapping status values

Status	Meaning
mapped_exact	Raw field is scientifically equivalent to canonical field
mapped_with_conversion	Equivalent after unit or representation conversion
mapped_conditional	Meaning depends on investigation documentation
unresolved	Requires subject-matter review
rejected	Field is not equivalent or is unusable

28. Validation checklist

Before a normalized dataset enters the prototype, confirm:

- Each test has a stable test identifier.
- Original filenames, column names, values, and units remain available.
- Pressure is absolute or explicitly marked otherwise.
- Gas percentage basis is known.
- Oxygen partial pressure is calculated only when appropriate.
- Airflow value specifies local/nominal and velocity/volume/mass basis.
- Material and geometry are separated.

- Ignition inputs and ignition outcomes are separated.
- `burn rate` has been disambiguated.
- Image-derived metrics include camera and threshold information.
- Time-series clocks are synchronized or their offsets are known.
- Missing values are not converted to zeros.
- Derived values identify their formula or model.
- Comparability labels have reasons.
- Every dashboard insight links back to source evidence.

29. Research gaps captured by the schema

The ontology should make these gaps visible rather than silently filling them:

- Sparse partial-gravity data for lunar and Martian conditions
- Limited realistic spacecraft materials compared with PMMA and cellulose
- Limited cross-experiment imaging calibration
- Missing low-speed in-situ flow measurements
- Scale differences between small test chambers and realistic habitats
- Under-studied smoke, toxicity, and post-fire cleanup
- Limited suppression and reignition evidence for practical configurations
- Incomplete historical coverage, particularly older experiments
- Uncertain transfer from Earth flammability tests to reduced gravity
- Sparse independent replicates for some parameter combinations

30. Research sources and local inputs

Local catalog used

- `flame_in_freefall_data_catalog.xlsx` - 24 PSI investigations, experimental-table links, file manifest, and NTRS references.
- `flame_in_freefall_data/experimental_tables.csv` - links to 22 PSI experimental tables.
- `flame_in_freefall_data/investigations.csv` - investigation objectives, platforms, DOI links, and dataset sizes.
- `flame_in_freefall_data/psi_files_manifest.csv` - analyzed data and documentation manifest.
- `flame_in_freefall_data/ntrs_reports.csv` - selected NASA Technical Reports Server references.

Authoritative external references

1. NASA Physical Sciences Informatics repository: https://psi.nasa.gov/physci/repo
2. NASA PSI investigations by research area: https://www.nasa.gov/physical-sciences-informatics-psi/psi-investigations-by-research-area/
3. NASA combustion and fire-safety overview: https://www.nasa.gov/missions/station/iss-research/studying-combustion-and-fire-safety/
4. NASA-STD-6001, Flammability, Offgassing, and Compatibility Requirements and Test Procedures: https://standards.nasa.gov/standard/NASA/NASA-STD-6001
5. NASA Technical Reports Server: https://ntrs.nasa.gov/

6. Challenges and Research Needs for Micro- and Partial-Gravity Fires: https://smd-cms.nasa.gov/wp-content/uploads/2023/05/60_afb22b1e538432728848e83bb3ade8c2_LiaoYaTingT.pdf
7. NIST Fire Dynamics Simulator and Smokeview: https://pages.nist.gov/fds/

31. Maintenance guidance

When a new dataset is added:

1. Add any new raw aliases to the synonym map.
2. Reuse existing canonical parameters whenever the physical meaning matches.
3. Add a new canonical parameter only when the scientific meaning is genuinely different.
4. Define type, role, preferred unit, valid range, and provenance requirements.
5. Record which investigation introduced the new parameter.
6. Update the dataset-family coverage map.
7. Regenerate the PDF and review the rendered pages.

The Markdown file is the editable source of truth. The PDF is a generated publication artifact.